

To: The Editors-in-Chief
Computer Physics Communications (CPC)
Elsevier Science
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Subject: Submission of Original Software and Computational Physics Manuscript
Title: *LeanFlow: A Certified Neuro-Symbolic Architecture for Stiff Moduli Dynamics, Topological T-Duality, and Closed-Loop Anomaly Verification in String Cosmology*

Dear Editors of Computer Physics Communications,

We submit our original research manuscript for consideration as a Regular Article in *Computer Physics Communications*.

High-energy cosmological systems and string compactifications involve stiff, multi-scale systems of partial and ordinary differential equations. When scalar oscillation frequencies ($\omega \sim 10^3 H$) vastly exceed the cosmic expansion rate, classical explicit integrators experience catastrophic step-size collapse, while standard implicit solvers can drift into unphysically singular regions of moduli space. Concurrently, while interactive theorem provers (ITPs) like Lean 4 offer formal machine-checked proof certificates down to a minimal kernel, their interactive elaboration engines incur prohibitive interpretation latency (tens of milliseconds), historically precluding their use inside stiff numerical PDE integration loops.

In this work, we introduce **LeanFlow**: a production-grade neuro-symbolic framework that bridges the compiled **rusty-SUNDIALS** stiff integration suite (CVODE and IDA BDF algorithms) with automated formal verification in Lean 4.

The primary computational and algorithmic contributions of our work are:

1. **Dual-Tier Execution Architecture:** LeanFlow decouples formal invariant verification from numerical integration:
 - *Inner Loop (Microsecond Stepping):* Mathematical invariants verified in Lean 4—including metric positivity on the upper half-plane ($\tau_{\text{im}} > 0$), the Weak Energy Condition bound ($w \geq -1$), and Buscher reflection projectors—are compiled Ahead-Of-Time (AOT) into zero-overhead Rust SIMD contract assertions. Incurring negligible runtime overhead ($\approx 12\text{--}18\text{ ns}$ per residual evaluation), these inline locks enforce physical admissibility at zero penalty to the stiff BDF stepping performance (0.12 ms step latency).
 - *Outer Loop (Asynchronous Dynamic Batch Gate):* Full Lean 4 proof kernel verification is invoked asynchronously via non-blocking Unix domain socket IPC / C-FFI ($\approx 45\text{ ms}$ batch latency) at macroscopic simulation milestones, checkpoint boundaries, and topological defect extraction sweeps.
2. **Native Duality Engine:** The integrator natively embeds continuous Buscher symmetry ($R \cdot \tilde{R} = \alpha'$) and modular group $SL(2, \mathbb{Z})$ invariance as an operational numerical section condition. By co-simulating dual geometric frames and folding trajectories across the fundamental domain boundary $\mathcal{F}$, LeanFlow prevents coordinate singularities near self-dual points ($R \approx \sqrt{\alpha'}$) without metric breakdown ($\min(y) = 0.289$).
3. **Closed-Loop Neuro-Symbolic Triad:** We achieve an unbroken circuit linking numerical stochastic Langevin simulations to Topological Data Analysis (TDA) Mapper (extracting 184 nodes, 493 edges, $\beta_1 = 318$), partitioning defect configurations into string-theoretic

equivalence classes, and certifying Ramond-Ramond tadpole cancellation ($\sum Q_i = 0$) in the Lean 4 kernel with zero `sorry` axioms.

4. **Rigorous Benchmark Performance:** On identical non-linear cosmological equations (Einstein-Klein-Gordon moduli flow, Mukhanov-Sasaki perturbations, and Symmetron screening BVP), LeanFlow achieves a **1,520** $\times$ speedup over standard Python/SciPy BVP baselines and a **7.1** $\times$ speedup over standard C++ SUNDIALS/CVODE. Containerized for serverless spot GPU/TPU execution with scale-to-zero capability ($\text{min} = 0$), it reduces computational costs by 99.1% relative to dedicated HPC clusters.

All code, proofs, and simulation pipelines are open-source and verified against an automated test suite (27/27 pytest tests passing, 32/32 Lean jobs building with zero `sorry`).

We believe LeanFlow represents a transformative paradigm for verifiable computational physics and will be of immediate value to the readers of CPC.

Thank you for your consideration.

Sincerely,

The Authors

SocrateAI Research Initiative